

UNIX Shell Programming

Course: 24CSAEC39 / CSAEC310 · **Semester:** III · **Max Marks:** 50 · **Duration:** 2 Hrs

Built from: 3 past papers — April 2023 (CSAEC310) · August 2023 backlog (CSAEC410) · March 2024 (CSAEC310)

Filtered against: the current 5-unit syllabus. Every out-of-syllabus question has been stripped out.

⚠ READ THIS FIRST — your syllabus is NOT the syllabus these papers were set from

All three past papers follow an **older unit split**. Comparing them line-by-line against your syllabus, two units have moved. If you revise straight from these papers you will walk in having prepared the wrong Unit V entirely.

Your syllabus	Where the past papers put it	Consequence
Unit IV — Filters + Regular Expressions using grep	Split across PYQ Unit IV (filters) and PYQ Unit V (grep/regex)	Two old units now compress into one . Unit IV is the densest unit on your paper.
Unit V — Shell Scripts (read , if , test , case , expr , while , for , set / shift)	Nowhere. Zero coverage in all three papers.	8 Part-B marks + ~2 Part-A marks cannot be predicted from PYQs at all. Section 10 builds these from the syllabus instead.
Unit II — process utilities	Job control (bg / fg , job states, signals) sat in PYQ Unit III	Expect job control to shift into your Unit II . Prepare it under Unit II, not III.

Bottom line: the PYQs reliably predict roughly **40 of 50 marks** (Units I–IV). The remaining 10 marks are Unit V and must be studied cold. Section 11 does the arithmetic.

1 · Exam pattern, decoded

Identical across all three papers — treat this as fixed.

Section	Marks	Choice	Structure
PART A	$10 \times 1 = 10$	None — answer all 10	MCQ. Exactly 2 questions per unit, in unit order: Q1–2 = Unit I, Q3–4 = Unit II, Q5–6 = Unit III, Q7–8 = Unit IV, Q9–10 = Unit V. Confirmed in all 3 papers, no exceptions.
PART B	$5 \times 8 = 40$	Internal choice — one full question from each unit	Each unit offers two questions. Unit I → Q1 or Q2. Unit II → Q3 or Q4. Unit III → Q5 or Q6. Unit IV → Q7 or Q8. Unit V → Q9 or Q10. You must take the whole question (both a and b) — no mixing a from one with b from the other.

Sub-part mark splits (every Part-B question totals 8)

Split	Frequency	What it means for you
4 + 4	13 of 30 questions	Most common. Two equally-weighted parts, usually two separate topics.
5 + 3	11 of 30	(a) is the diagram / long theory, (b) is a short command explanation or a "write the commands" list.
3 + 5	5 of 30	Appeared only in March 2024 — the newest paper. Rising.
6 + 2	1 of 30	Rare outlier (Aug 2023 only).

Rule of thumb: a 4–5 mark part wants a **labelled diagram + explanation**. A 2–3 mark part wants **syntax + one worked example**. Never write a diagram-less answer for a 5-marker.

2 · Ruled out — do NOT study these

These appeared in the past papers but are **absent from your syllabus**. Cutting them saves you real time.

Topic	Where it appeared	Why it's out
sed (all of it)	Aug 2023 Q10(a) 4m — sed addresses; Q10(b) 4m — insert/append/change/delete; Part A Q10 — <code>sed 's/cat/dog/g'</code>	Your syllabus never mentions <code>sed</code> . Unit IV stops at "Regular Expressions using grep". Skip entirely.
awk	March 2024 Q8(b) 5m — built-in variables in awk	Not in any unit. Unit IV lists filters by name and <code>awk</code> is not among them. Skip entirely.
pr command	Aug 2023 Part A Q7 — preparing a file for printing	Not in the Unit I command inventory, not named in Unit IV. Borderline — 1 mark of risk , ignore unless you have spare time.

Careful with the sed/awk cut: that is **9 marks** of the March 2024 + Aug 2023 papers deleted. The examiner must refill that space from your syllabus — and the only thing that expanded is **Unit V shell scripting**. This is independent evidence that Unit V will be examined properly, not token-tested.

3 · TIER 1 — Locked. Same question, same shape, expect it verbatim.

These recur across papers with near-identical wording. Memorise the answer *as a unit* — diagram included.

TIER 1

Illustrate / explain the various stages of a UNIX session with a neat diagram.

Unit: II Marks: 4–5 Section: Part B, Q3(a) Pairs with: a process question in 3(b)

Evidence — 3 of 3 papers, always Q3(a): Apr 2023 "Analyze Unix session flow with a neat diagram" (5m) · Aug 2023 "With a neat diagram, explain the different stages of UNIX session" (4m) · Mar 2024 "Illustrate the various stages of a UNIX session with a neat diagram" (4m). **The single most repeated question in the entire dataset.** Wording barely changes.

TIER 1

Explain the three standard streams / standard redirection files of UNIX.

Unit: III Marks: 3–5 Section: Part B, Q5(a) Pairs with: tee, or variables, in 5(b)

Evidence — 3 of 3 papers, always Q5(a): Apr 2023 "Explain in detail three standard streams of UNIX" (5m) · Aug 2023 "What is Redirection? Explain any 2 redirecting techniques with a neat diagram" (5m) · Mar 2024 "Explain the three standard redirection files with respect to UNIX OS" (3m). Answer covers `stdin(0)/stdout(1)/stderr(2) + > >> < 2>`.

TIER 1

Explain the five categories of operators used in regular expressions.

Unit: IV (was PYQ Unit V) Marks: 5 Section: Part B, Q7 or Q8 Pairs with: grep operation / atoms

Evidence — 2 of 3 papers, 5 marks both times: Aug 2023 Q9(a) "Explain 5 different categories of regular expression operators" (5m) · Mar 2024 Q10(b) "Describe the five categories of operators commonly utilized in regular expressions" (5m). The phrase **"five categories"** is the examiner's fixed formula.

TIER 1

What is grep? Explain the grep operation with a neat diagram / flow chart.

Unit: IV (was PYQ Unit V) Marks: 3–5 Section: Part B, Q7 or Q8 Pairs with: regex operators / atoms

Evidence — 2 of 3 papers: Aug 2023 Q9(b) "What is grep? Explain grep operation with a neat diagram" (3m) · Mar 2024 Q9(a) "What is grep? Explain grep operation using a simple flow chart" (5m). Note grep also carries a Part-A MCQ in **all three** papers.

TIER 1 Explain the `tr` command with its options / the various ways of using `tr` .

Unit: IV Marks: 4–5 Section: Part B, Q8(a) Pairs with: `uniq` in 8(b)

Evidence — 2 of 3 papers, both as Q8(a): Apr 2023 "Explain with examples the various ways of using the 'tr' command" (5m) · Aug 2023 "Explain 'tr' command with its options" (4m). Syllabus names it explicitly: "Translating Characters".

TIER 1 What are file permissions? Explain `chmod` using both the absolute (octal) and relative (symbolic) methods.

Unit: I Marks: 3–4 Section: Part B, Q2 Pairs with: command explanations in the other half of Q2

Evidence — 2 of 3 in Part B + 3 of 3 in Part A: Apr 2023 Q2(a) "Explain two methods to change permission of files" (4m) · Mar 2024 Q2(b) "chmod ... absolute and relative methods" (3m). Part A: Apr 2023 Q2 (default permissions), Aug 2023 Q2 (octal 744/755), Mar 2024 Q2 (relative-mode components). Syllabus: "security by file permissions".

TIER 1 Explain the `du` command with its different options.

Unit: II Marks: 3–4 Section: Part B, Q4 Pairs with: text-processing commands

Evidence — 2 of 3 papers: Apr 2023 Q4(b) "Explain 'du' command with options" (3m) · Aug 2023 Q4(a) "Analyze the working of 'du' command with different options" (4m). Syllabus: "disk utilities". Add `df` — it is the natural companion and has not yet been asked.

TIER 1 Explain the `tee` command / demonstrate `tee` integrated into a pipeline, with a diagram.

Unit: III Marks: 4–5 Section: Part B, Q5(b) or Q6(a) Pairs with: redirection / alias

Evidence — 2 of 3 in Part B + 1 Part-A MCQ: Aug 2023 Q6(a) "Demonstrate the working of 'tee' command with necessary diagram" (4m) · Mar 2024 Q5(b) "Explain how the tee command can be integrated into a pipeline" (5m) · Aug 2023 Part A Q5 (tee MCQ). Syllabus names "Tee Command" explicitly.

TIER 1 Name and explain any three / four UNIX commands used for text processing.

Unit: II Marks: 3–4 Section: Part B, Q4 Pairs with: `du` or networking commands

Evidence — 2 of 3 papers: Aug 2023 Q4(b) "Name and explain any four UNIX commands used for text processing" (4m) · Mar 2024 Q4(a) "List and explain any three UNIX commands used for text processing" (3m). Only the *count* changes — prepare four and you cover both.

TIER 1 Illustrate the different job states in UNIX with a neat diagram. / Explain `bg` and `fg` .

Unit: II (*moved — was PYQ Unit III*) Marks: 3–4 Section: Part B Pairs with: process utilities / signals

Evidence — 3 of 3 papers: Apr 2023 Q6(a) "Illustrate the job states in UNIX with a flow diagram" (5m) · Aug 2023 Q5(b) "Illustrate different job states with a neat diagram" (3m) · Mar 2024 Q6(a)+(b) SIGSTOP / bg-fg (4+4m). **Position warning:** your syllabus files this under *process utilities* (Unit II), not Standard Streams (Unit III). Prepare it, but expect it one unit earlier.

4 · TIER 2 — Locked topic, but the values / command list WILL change

These appear every single time, but the examiner swaps the specific commands. Don't memorise one answer — **learn the whole command family** so any combination is answerable.

TIER 2 Explain the following commands with syntax, options and an example: (*list varies*)

Unit: I Marks: 4–5 Section: Part B, Q1(b) or Q2(a) Pairs with: chmod / architecture

Evidence — 3 of 3, list changes every time: Apr 2023 Q1(b) asked `wc`, `cp`, `who` · Aug 2023 Q1(b) "any four basic UNIX commands" · Mar 2024 Q2(a) asked `echo`, `ls`, `who`, `passwd`, `date`.

Safe prep set (covers all three + your syllabus inventory): `echo` `printf` `ls` `who` `date` `passwd` `uname` `pwd` `cd` `mkdir` `rmdir` `cp` `mv` `rm` `cat` `more` `wc` `man` `script` `stty` `od` `lp` `tar` `gzip`. Learn syntax + 2 options + 1 example each — that is the exact format the marks are awarded in.

TIER 2 UNIX architecture / kernel–shell relationship — explain with a neat labelled diagram.

Unit: I Marks: 4–5 Section: Part B, Q1(a) Pairs with: features of UNIX in 1(b)

Evidence — 2 of 3, same diagram, different framing: Apr 2023 "Illustrate the layered UNIX architecture with a neat diagram" (5m) · Mar 2024 "Explain with a figure, the kernel and shell relationship" (4m). One concentric diagram (hardware → kernel → shell → applications) answers both wordings. Aug 2023 used the Q1(a) slot for vi-editor modes instead — see Tier 3.

TIER 2 Explain the following filter commands: `cut`, `paste`, `head`, `tail` (*subset varies*)

Unit: IV Marks: 3–4 Section: Part B, Q7 Pairs with: `wc` or a "write the commands" list

Evidence — 3 of 3, subset changes: Apr 2023 Q8(b) tail/head/paste (3m) · Aug 2023 Q7(b) head -2, tail -2, middle-of-file (3m) · Mar 2024 Q7(a) cut/paste/head/tail (4m). Syllabus wording — "Display Beginning and End of Files, Cut and Paste" — guarantees this cluster.

TIER 2 Process utilities — types of processes, initialization & termination, `ps`, and priority with `nice` / `renice`.

Unit: II Marks: 4–5 Section: Part B, Q3(b) or Q4(a) Pairs with: the UNIX session diagram in 3(a)

Evidence — 3 of 3, angle changes: Apr 2023 Q4(a) "Explain process utilities in detail" (5m) · Aug 2023 Q3(b) "types of process ... initialization and termination" (4m) · Mar 2024 Q3(b) "process priorities ... nice and renice" (4m). Cover all three angles — they rotate.

TIER 2 Explain `grep` with its options / how `grep` searches file contents / `grep` vs `egrep` vs `fgrep`.

Unit: IV Marks: 3–4 Section: Part B, Q7 or Q8 Pairs with: regex operators, grep flow chart

Evidence — 3 of 3 in Part B + 3 of 3 in Part A: Apr 2023 Q10(a) "grep command with options" (4m) & Q10(b) "how grep can be used to search file contents" (4m) · Mar 2024 Q10(a) "Differentiate between grep, egrep, fgrep" (3m). Options to know cold: `-i` `-v` `-c` `-n` `-l` `-e` `-E` `-f` `-X`.

TIER 2 Write the UNIX commands to perform the following operations: (*3–4 practical tasks, always different*)

Unit: I, II or IV Marks: 3–4 Section: Part B, the (b) part Pairs with: whatever theory sits in the (a) part

Evidence — 2 of 3 papers, four separate instances: Apr 2023 Q1(b) `wc/cp/who` · Q3(b) `sort -u`, `uniq -d`, `cmp` · Q7(b) `head -5`, `cut -f1,3`, `paste`, `wc -l` · Aug 2023 Q7(b) `head -2`, `tail -2`, middle lines. Tasks never repeat — but they are always one-liners drawn from the syllabus command list. Practice writing, not reciting.

5 · TIER 3 — High probability, single-paper precedent or syllabus-mandated

TIER 3 Illustrate the working of the `vi` editor with its different modes.

Unit: I Marks: 4 Section: Part B, Q1(a) Pairs with: basic commands in 1(b)

Aug 2023 Q1(a), 4m. Named explicitly in your syllabus ("vi editor") yet asked only once in three papers — and not at all in the most recent. Three modes: Command / Insert / ex (Escape). Overdue and cheap. Do it.

TIER 3 List and explain the salient features of the UNIX operating system.

Unit: I **Marks:** 4 **Section:** Part B, Q1(b) **Pairs with:** the architecture diagram in 1(a)

Mar 2024 Q1(b), 4m — the most recent paper. Syllabus: "Features of UNIX". Multi-user, multitasking, portability, hierarchical file system, security, shell, pipes/filters, communication. **Pairs directly with the Tier-2 architecture diagram — learn them as one answer.**

TIER 3 Explain the `uniq` command with its different options.

Unit: IV **Marks:** 4 **Section:** Part B, Q8(b) **Pairs with:** `tr` in 8(a)

Aug 2023 Q8(b), 4m — plus **two separate Part-A MCQs** (Mar 2024 Q3 `-d`, Aug 2023 Q8 `-u`). Syllabus: "Files with Duplicate Lines". Options: `-u -d -c -i`. Remember it needs **sorted input**.

TIER 3 What is an alias? Explain creating and removing aliases with examples.

Unit: III **Marks:** 3–4 **Section:** Part B, Q6(b) **Pairs with:** `tee` / command substitution / variables

Aug 2023 Q6(b), 4m. Syllabus names "Aliases" explicitly. `alias`, `unalias`, and the lab component is literally "customization of Linux shell environment" — which is aliases + env variables.

TIER 3 Explain the types of variables in UNIX with examples. / Explain environment variables.

Unit: III **Marks:** 3–4 **Section:** Part B, Q5(b) or Q6(b) **Pairs with:** streams / redirection

Apr 2023 Q5(b), 3m. Syllabus lists **both** "Variables" and "Environment Variables" as separate items — two syllabus entries, one question. Local vs environment vs shell; `PATH HOME SHELL PS1 PS2 LOGNAME`; `set`, `export`, `env`.

TIER 3 Explain the need for command substitution with an example.

Unit: III **Marks:** 3 **Section:** Part B, Q6(b) **Pairs with:** job control / alias

Apr 2023 Q6(b), 3m. Syllabus: "Command Substitution". Backticks vs `$( )`, with `echo "Today is $(date)"`. **Small topic, ~10 minutes to learn, 3 guaranteed-shaped marks.**

TIER 3 Explain the commands used to compare two files (`cmp`, `comm`, `diff`).

Unit: II / IV **Marks:** 3–4 **Section:** Part B, Q7(a) or Q8(a) **Pairs with:** filter commands

Apr 2023 Q7(a) "any two commands to compare two files" (4m) · Mar 2024 Q8(a) "diff with `-c`, `-u`, `-i`" (3m) · Part A: Aug 2023 Q4, Mar 2024 Q8 (`cmp` exit status = 0 when identical). **2 of 3 in Part B, 2 of 3 in Part A.** Falls under "UNIX utilities" in Unit II.

TIER 3 Explain the command used to concatenate files (`cat`) with its options.

Unit: IV **Marks:** 4–5 **Section:** Part B, Q7(a) **Pairs with:** `head`/`tail` practical commands

Aug 2023 Q7(a), 5m. Syllabus names it as its own line item: "Concatenating Files". `cat -n -v -s -e`, plus creating and appending files with `>` and `>>`.

TIER 3 Briefly explain the networking commands: `ping`, `host`, `tracert`, `netstat`, `route`.

Unit: II **Marks:** 5 **Section:** Part B, Q4(b) **Pairs with:** text-processing commands in 4(a)

Mar 2024 Q4(b), 5m — most recent paper. Part A: Apr 2023 Q4 (`ping`), Mar 2024 Q4 (`route`). Syllabus: "networking commands". Also know `ftp`, `telnet`, `ssh`, `ifconfig`, `finger`.

TIER 3 Explain the different wildcards / shell metacharacters used in UNIX with examples.

Unit: I or III Marks: 4 Section: Part B, Q2(b) Pairs with: file permissions in 2(a)

Apr 2023 Q2(b), 4m. Sits under "file handling utilities" (Unit I) or "Quotes / Command Execution" (Unit III). * ? [] [!] plus escaping and quoting — and quoting is a named syllabus item that has never been asked. See Section 6.

6 · OVERDUE — in your syllabus, never yet asked in Part B

Every one of these is named in your syllabus but has **zero Part-B appearances** across three papers. Examiners rotate; these are statistically the next to surface. Each is small — this section is cheap insurance.

Topic	Unit	Syllabus wording	What to prepare
Sorting — sort	IV	"Sorting."	A whole named syllabus line, and only ever a Part-A MCQ (Apr 2023 Q8, delimiter). Options -n -r -k -t -u -m . Highest-value item in this table.
Quoting & escaping	III	"Quotes"	Single vs double vs backslash; what each protects. Never asked in Part B. Pairs naturally with wildcards.
Backup utilities — tar , gzip	I & II	"tar, gzip" + "backup utilities"	Listed in two units, only ever one Part-A MCQ (Aug 2023 Q3, -c). tar -cvf/-xvf/-tvf , gzip / gunzip / zcat .
Command execution modes	III	"Command Execution"	Sequential ; , background & , grouping () , && / . Overlaps Unit V — double-dip .
script , stty , od , PATH , man	I	Named in the Unit I command list	Never asked. Likely as one item inside a "explain the following commands" list — see Tier 2.
UNIX file system structure	I	"Introduction to UNIX file system"	The / hierarchy tree (/bin /etc /dev /usr /tmp /home), file types, inodes, absolute vs relative paths. A named syllabus heading with no Part-B precedent — and it is a diagram question, which is exactly what 4–5 markers want.

7 · PART A — the ten 1-mark MCQs

Two per unit, in order, no choice. Ranked by how often the topic has carried a Part-A slot.

Slot	Unit	Predicted topic	Hit rate	The fact to memorise
Q1	I	What the shell is / shell's role	2 of 3	Shell is both a scripting language and an interpreter ; it is the interface between user and kernel.
Q2	I	chmod / permissions	3 of 3	Default file perms rw- rw- rw- (666 before umask); 755 =rwxr-xr-x ; 744 =rwxr--r-- ; relative mode has 3 components (who + operator + permission).
Q3	II	Process / ps / tar	2 of 3	ps shows process attributes; tar -c creates an archive, -x extracts, -t lists.
Q4	II	Networking or file-comparison command	3 of 3	ping tests reachability; route manipulates the TCP/IP routing table; diff / cmp / comm compare files.
Q5	III	Pipe / redirection / tee symbol	3 of 3	 connects streams; & pushes to background; tee copies stdin to stdout <i>and</i> a file.

Slot	Unit	Predicted topic	Hit rate	The fact to memorise
Q6	III	Redirection operators or job control	3 of 3	Output redirection = <code> </code> , <code>></code> and <code>>></code> (the "all of these" option is usually right); <code>Ctrl+Z</code> suspends a foreground job.
Q7	IV	cut options	2 of 3	<code>-c</code> = characters, <code>-f</code> = fields, <code>-d</code> = delimiter. <code>cut -c -3</code> means columns 1–3.
Q8	IV	uniq options / sort / exit status	3 of 3	<code>uniq -d</code> = repeated lines only, <code>-u</code> = non-repeated only, <code>-c</code> = count. <code>sort</code> default delimiter = one or more spaces . <code>cmp</code> exit status = 0 if identical.
Q9	V*	Shell scripting basics (<i>new</i>)	—	No PYQ precedent. Expect: <code>\$?</code> = exit status of last command; 0 = success; <code>\$#</code> = number of arguments; <code>\$0</code> = script name.
Q10	V*	Shell scripting constructs (<i>new</i>)	—	No PYQ precedent. Expect: <code>shift</code> moves positional parameters left; <code>esac</code> closes <code>case</code> ; <code>fi</code> closes <code>if</code> ; <code>-eq</code> is numeric, <code>=</code> is string.

* **Q9/Q10 warning:** in the old papers these two slots were **grep and regular expressions** (3 of 3 papers). Under your syllabus grep moved to Unit IV, so Q9/Q10 should become shell scripting. **Prepare both** — if the paper still uses the old mapping you get grep MCQs, and grep is Tier 1 anyway. Either way you are covered.

8 · Which questions appear TOGETHER — your choice map

This is the section that wins marks. Each unit gives you **two** questions and you take one **whole**. Below is what historically bundles with what, so you can decide *before* the exam which option you are taking.

UNIT I — take Q1 or Q2

Q1 — the "diagram + theory" question

- (a) Layered UNIX architecture diagram [Apr 23, 5m]
- (a) vi editor modes [Aug 23, 4m]
- (a) Kernel–shell relationship figure [Mar 24, 4m]
- (b) Commands: wc, cp, who [Apr 23, 3m]
- (b) Any four basic commands [Aug 23, 4m]
- (b) Salient features of UNIX [Mar 24, 4m]

Q2 — the "commands + permissions" question

- (a) Two methods to change permissions [Apr 23, 4m]
- (a) cp command with options [Aug 23, 6m]
- (a) Explain echo/ls/who/passwd/date [Mar 24, 5m]
- (b) Wildcards [Apr 23, 4m]
- (b) Rename a file [Aug 23, 2m]
- (b) chmod absolute + relative [Mar 24, 3m]

VERDICT: Q1 always = a structural diagram + descriptive theory. Q2 always = command explanations + file permissions. **Prepare the architecture diagram + features (Q1) AND chmod + the command list (Q2)** — that is four small topics and it makes Unit I choice-proof.

UNIT II — take Q3 or Q4

Q3 — "UNIX session + processes"

- (a) UNIX session stages, neat diagram — **ALL 3 PAPERS** [5m, 4m, 4m]
- (b) Commands: sort -u, uniq -d, cmp [Apr 23, 3m]
- (b) Types of process, init & termination [Aug 23, 4m]
- (b) Process priorities, nice & renice [Mar 24, 4m]

Q4 — "disk + text + network utilities"

- (a) Process utilities in detail [Apr 23, 5m]
- (a) du command with options [Aug 23, 4m]
- (a) Three text-processing commands [Mar 24, 3m]
- (b) du command with options [Apr 23, 3m]
- (b) Four text-processing commands [Aug 23, 4m]
- (b) Networking commands [Mar 24, 5m]

VERDICT — ANSWER Q3. The UNIX session diagram is **3 for 3 in the Q3(a) slot** — the most reliable 4–5 marks on the entire paper. It always pairs with a **process** question in 3(b). So: learn the session diagram + process types + nice/renice, and Unit II is 8/8 without touching du or networking.

UNIT III — take Q5 or Q6

Q5 — "streams & redirection"

(a) **Three standard streams / redirection** — ALL 3 PAPERS [5m, 5m, 3m]

(b) Types of variables in UNIX [Apr 23, 3m]

(b) Job states diagram [Aug 23, 3m]

(b) tee in a pipeline [Mar 24, 5m]

Q6 — "job control + shell features"

(a) Job states flow diagram [Apr 23, 5m]

(a) tee command with diagram [Aug 23, 4m]

(a) SIGSTOP / job control signals [Mar 24, 4m]

(b) Command substitution [Apr 23, 3m]

(b) Alias — create and remove [Aug 23, 4m]

(b) bg and fg commands [Mar 24, 4m]

VERDICT — ANSWER Q5. Standard streams/redirection is **3 for 3 in the Q5(a) slot**. The (b) partner rotates between **variables / job states / tee** — all three are Tier 1 or Tier 3 here, so prepare all three (they are 15 minutes each) and Q5 is locked at 8/8. **Note:** under your syllabus, job control likely migrates to Unit II — so Q5(b) will most likely be **tee or variables**.

UNIT IV — take Q7 or Q8 ▲ **restructured unit**

Your Unit IV absorbs **both** old PYQ Unit IV (filters) **and** the grep/regex half of old PYQ Unit V. Two scenarios — prepare for both, they overlap heavily.

Q7 — most likely "filters"

(a) Two commands to compare files [Apr 23, 4m]

(a) cat / concatenating files [Aug 23, 5m]

(a) cut, paste, head, tail [Mar 24, 4m]

(b) Write commands: head -5, cut -f1,3, paste, wc -l [Apr 23, 4m]

(b) Write commands: head -2, tail -2, middle [Aug 23, 3m]

(b) wc — lines, words, characters [Mar 24, 4m]

Q8 — most likely "tr / uniq / grep + regex"

(a) tr — various ways of using [Apr 23, 5m]

(a) tr with options [Aug 23, 4m]

(a) ~~diff -e -u -i~~ [Mar 24, 3m]

(b) tail, head, paste [Apr 23, 3m]

(b) uniq with options [Aug 23, 4m]

(b) ~~awk built-in variables~~ — **OUT OF SYLLABUS**

+ **absorbed from old Unit V:** five categories of regex operators [5m]

· grep operation flow chart [5m] · grep vs egrep vs fgrep [3m] · atoms and operators [5m]

VERDICT: Q7 = the **filter family** (cat, head, tail, cut, paste, wc, sort). Q8 = **tr, uniq, and the grep/regex block**. Because awk and sed are cut, Q8 has a **5-mark hole** that grep/regex will almost certainly fill. **Prepare the filter family for Q7 and tr + uniq + "five categories of regex operators" for Q8** — that covers both branches.

UNIT V — take Q9 or Q10 ▲ **zero PYQ precedent — see Section 9**

VERDICT: No past paper can help you here. The old Q9/Q10 slots held grep and regular expressions, which have moved to your Unit IV. Section 9 predicts this unit from the syllabus text itself.

9 · UNIT V — Shell Scripts. The blind spot.

This unit is worth 8 Part-B marks + roughly 2 Part-A marks = 10 marks, or 20% of the paper — and not one of your three past papers contains a single shell-scripting question. Every prediction below is derived from the syllabus topic list and the paper's own 8-mark two-part structure, not from precedent. Treat this section as the difference between scraping 62 and comfortably clearing 80.

Your syllabus lists Unit V topics in this exact order — the examiner will almost certainly walk the same order when splitting two questions:

`read` → command line arguments → `exit` & exit status → `&&` `||` → conditional execution → `if` → `test` and `[ ]` → `case` → `expr` → `$0` → `while` → `for` → `set` and `shift`

That is 13 topics across 2 questions × 2 parts = 4 slots. The natural split falls at the `case` / `expr` boundary:

PREDICTED Q9 — the "input, arguments and conditionals" question

Unit: V **Marks:** 8 (as 4+4 or 5+3)

(a) Most likely [4–5m] — one of:

- Explain `test` and `[ ]` for evaluating expressions. Give the file / string / numeric operator tables.
- Explain the `if` conditional with its forms (`if-fi` , `if-else-fi` , `if-elif-else-fi`) with a shell script example.
- Explain how to make a script interactive using `read` , with an example.

(b) Most likely [3–4m] — one of:

- Explain command line arguments / positional parameters: `$0` `$1...``$9` `$#` `$*` `$@` `$?`
- Explain `exit` and the exit status of a command. Why is `0` = success?
- Explain the logical operators `&&` and `||` with examples.

PREDICTED Q10 — the "case, computation and loops" question

Unit: V **Marks:** 8 (as 4+4 or 3+5)

(a) Most likely [4–5m] — one of:

- Explain the `case` conditional with syntax and a menu-driven shell script example.
- Explain `while` looping and `for` looping with a list, with one example each.
- Explain `expr` for computation and string handling (`expr length` , `expr substr` , `expr index`).

(b) Most likely [3–4m] — one of:

- Explain `set` and `shift` for manipulating positional parameters, with an example.
- Explain `$0` — calling a script by different names — and why it is useful.
- Write a shell script to... (see the drill list below).

The "write a shell script to..." drill list

The lab component is "Linux commands execution related to shell scripts", so at least one part will ask you to **write** a script. These are the standard first-course programs — each maps to a named syllabus construct. Write out all ten by hand once.

Program	Construct it tests
Read two numbers and print the larger	<code>read</code> + <code>if</code> + <code>test</code>
Check whether a number is even or odd	<code>expr</code> modulo + <code>if</code>
Find the largest of three numbers	nested <code>if-elif-else</code>
Menu-driven calculator (+ − × ÷)	<code>case</code> + <code>expr</code> + <code>while</code>
Factorial of a number	<code>while</code> + <code>expr</code>
Sum of digits / reverse a number	<code>while</code> + <code>expr</code>
Print the first N Fibonacci numbers	<code>for</code> or <code>while</code> + <code>expr</code>
Count the arguments passed and print each one	<code>\$#</code> + <code>shift</code> + <code>while</code>
Check if a file exists, and report its permissions	<code>test -f -r -w -x</code> + <code>if</code>
Print all files in a directory using a list	<code>for</code> looping with a list

Write them in the syllabus dialect, or you lose marks. Use `expr` for arithmetic — **not** `$(( ))`. Use `test` or `[ ]` — **not** `[[ ]]`. Your syllabus names `expr` and `test` explicitly and never introduces the alternatives, so the bash shortcuts read as out-of-syllabus to an examiner marking against a scheme.

10 · The scoring plan — what to study for what mark

Marks available, by what you actually revise. Read this as a ladder; each rung adds to the one above.

Rung	What you study	Time	Marks secured	Running total
1	Tier 1 only (Section 3) — 10 questions, all with diagrams	~3 hrs	~20	20 / 50 — fail
2	+ Tier 2 (Section 4) — the command families	+3 hrs	+10	30 / 50 — 60%
3	+ Part A facts (Section 7)	+1 hr	+7	37 / 50 — 74%
4	+ Unit V shell scripting (Section 9)	+3 hrs	+8	45 / 50 — 90%
5	+ Tier 3 & Overdue (Sections 5–6)	+2 hrs	+3	48 / 50 — safety margin

Your 60–70 target, answered precisely

Rungs 1–2 alone put you at 60% (30/50) on roughly six hours of work, entirely from repeated questions. **Rung 3 takes you to 74%.** That clears your target with margin, and none of it requires Unit V.

But do not stop there. Unit V is 8 Part-B marks you would be *guaranteed* to drop, and it is the one unit where the paper has no fallback — if you skip it, your realistic ceiling is 42/50 and your realistic floor, if a Tier-2 command list surprises you, is around 31. Three hours on Section 9 converts your worst case from "just passed" into "comfortably safe". **That is the single highest-return decision on this sheet.**

Sequence to revise in

- The four 3-of-3 diagrams first** — UNIX session stages (Unit II), standard streams/redirection (Unit III), UNIX architecture (Unit I), grep operation flow chart (Unit IV). Four diagrams, ~18 marks of exposure, and diagram marks are the easiest to earn cleanly.
- chmod, absolute and relative.** 3 of 3 in Part A, 2 of 3 in Part B. Nothing else on this paper has that density.
- The Unit I command list** with syntax + 2 options + 1 example each. This is the Tier-2 question that appears every year with a different list.
- tr, uniq, cut, paste, head, tail, cat, sort, wc** — the filter family. Unit IV is now the densest unit on your paper.
- Unit V shell scripts** — the 13 constructs, then hand-write the ten drill programs.
- Five categories of regex operators** — a fixed 5-mark formula, verbatim, twice.
- Mop up:** variables/environment variables, alias, command substitution, tee, du, process utilities, networking commands.

Method. Built by extracting and tabulating all 30 Part-B sub-questions and 30 Part-A MCQs across three papers, mapping each to your current syllabus, discarding out-of-syllabus items, then scoring by slot frequency and position. **Evidence keys:** Apr 2023 = CSAEC310, Sem III · Aug 2023 = CSAEC410, Sem IV backlog · Mar 2024 = CSAEC310, Sem III.

Caveat, stated plainly: three papers is a small sample, and two of the three predate your syllabus revision. Tier 1 rests on repetition that is genuinely strong; Unit V rests on the syllabus text alone and carries no precedent whatsoever. Prepare it as if it were unpredictable, because it is.